

Miscellaneous Problems

1. How many ways are there to color a 3×3 board using two colors, one color per square? Treat rotations and reflections of the board as different colorings.
2. How many ways are there to color a 3×3 board using three colors if (a) every row must contain all three colors; (b) every row and every column must contain all three colors? Treat rotations and reflections of the board as different colorings.
3. Sam is walking around a building in the form of a regular pentagon. The side length of the building is 30 meters. It is foggy and Sam can only see 3 meters around of him. What is the total area of the region that Sam can see on his walk?
4. Anna starts at point A , walks 1 mile north, then makes a 90 degree turn and walks 2 miles, then makes another 90 degree turn and walks 3 miles, then makes another 90 degree turn and walks 1 mile and stops at point B . Find all possible lengths AB .
5. If $a \oplus b = ab - a - b + 1$, find $1 \oplus (1 \oplus 2 + 2 \oplus 3 + 3 \oplus 4)$.
6. How many ways are there to color a cube using two colors, one color per face? Two colorings are considered identical if one can be obtained from the other by rotation in space.
7. Find the largest integer x such that $x^2 > 100$ and $x^3 < 64$.
8. Find the area of quadrilateral with vertices $(0, 1), (3, 5), (6, 2), (5, 0)$.
9. Find the area of a triangle formed by the lines $y = 3x + 4, y = 2x - 4, y = -2x + 4$.
10. Compute $\sqrt{9 - 6\sqrt{2}} + \sqrt{9 + 6\sqrt{2}}$.
11. A disk with radius 1 is rolled inside a circle with radius 2. Initially point P on the disk is touching the circle. What is the total distance point P will travel until it touches the circle again?
12. Find the coefficient of x^3 in the expansion $\left(x^2 + \frac{1}{x}\right)^{15}$.
13. Find 8 lattice points that are 5 units away from $(13, 71)$. A lattice point is a point with integer coordinates.
14. What is the probability that 3 randomly drawn cards from a standard deck with 52 cards are the same suit?
15. Find the number of terminal zeros of $100!$ when written in base 12.
16. Find the smallest positive integer k so that $n!$ cannot have k terminal zeros when written in base 9.
17. Let $p(x) = \prod_{n=0}^{100} (x - n)^n$.
 - (a) Find the number of digits in $p(37)$.
 - (b) Find the remainder when $p(600!)$ is divided by 97.
 - (c) Find the coefficient of the second term of $p(x)$ when completely expanded and written in standard form.

18. A king is on the bottom left cell of a 5×5 square. It can move up, right, or diagonally up-right 1 square. How many distinct paths are there from the bottom left to the top right cell?
19. Find the greatest common factor of the numbers 20727036 and 20727432.
20. Find the coefficient of x^{999} when $(x + 1)(x + 2)\dots(x + 1000)$ is fully expanded.
21. Find the area of a rhombus (quadrilateral with all equal sides), if one of its angles is twice another angle and one of its sides is 3 units long.
22. Find the last 4 digits of 101^{1000} .
23. Find a row in Pascal's triangle in which there are at least 8 terms divisible by 3.
24. Find the sum of the coefficients of $(2xy + z - 3)^{100}$.
25. Find the number of terminal zeros of the expression $\binom{1000}{500}$.
26. Two players play a win-by-2 game, that is, until one of them has 2 more points than the other. Anna is twice as likely to score a point than Bob. What is the probability that Anna will win the game?
27. Find a formula for the area of a regular octagon with side s .
28. If 5 men can dig 10 wells in 20 days, how many men are needed to dig 15 wells in 8 days?
29. Anna, Bob, Cat and Denis went to a store to buy candy. Anna wants to have at least as much candy as Bob and every child wants to have at least 2 pieces of candy. If they buy a total of 24 pieces of candy, in how many ways can they distribute them?
30. There are 3 pizzas, each pizza is cut into 8 slices. In how many ways can six kids divide the pizzas if everyone must get at least 1 slice but no one is allowed to get more than 5?